

Skip Module Hyperparameter Selection

1 Algorithm

Let F_t denote the grayscale frame at time t , converted from the downsampled input $\tilde{f}_t$. $\mathbf{M}_t \in \{0, 255\}$ is the binary foreground mask output, where a pixel is declared active if the absolute inter-frame intensity change exceeds threshold τ_d .

Algorithm 1 Frame Difference Detector (FrameDiffDet)

Require: Downsampled frames $\tilde{f}_t, f_{\text{prev}}$

Require: τ_d : pixel difference sensitivity threshold

Ensure: $\mathbf{M}_t \in \{0, 255\}$: binary foreground mask

1: Convert $\tilde{f}_t$ and f_{prev} to grayscale: F_t, F_{prev}

2: $\mathbf{M}_t(i, j) \leftarrow \begin{cases} 255 & |F_t(i, j) - F_{\text{prev}}(i, j)| > \tau_d \\ 0 & \text{otherwise} \end{cases}$

3: **return** $\mathbf{M}_t$

Let F_t denote the grayscale frame at time t , converted from the downsampled input $\tilde{f}_t$, and $\mathbf{K}_t \in [0, K_{\text{max}}]$ the per-pixel *accumulated motion mask*. Upon detection, $\mathbf{K}_t$ is incremented by ω and decays by δ each frame, bounded at K_{max} to prevent indefinite accumulation of stale evidence. A pixel is declared active only when $\mathbf{K}_t \geq \tau_m$, suppressing transient noise and single-frame flicker. The binary output $\mathbf{M}_t \in \{0, 255\}$ is consistent with the interface expected by the skip module $\mathcal{S}$.

Algorithm 2 Accumulation Motion Detector (AccMotionDet)

Require: Downsampled frames $\tilde{f}_t, f_{\text{prev}}$; accumulated mask $\mathbf{K}_{t-1}$ (initialized to 0)

Require: τ_d : pixel difference sensitivity, ω : motion increment, τ_m : activation threshold, K_{max} : accumulation cap, δ : decay rate

Ensure: $\mathbf{M}_t \in \{0, 255\}$: binary foreground mask; $\mathbf{K}_t$: updated accumulated mask

- 1: Convert $\tilde{f}_t$ and f_{prev} to grayscale: F_t, F_{prev}
 - 2: $\mathbf{D}_t \leftarrow |F_t - F_{\text{prev}}|$ ▷ Absolute inter-frame difference
 - 3: $\mathbf{\Delta}_t(i, j) \leftarrow \begin{cases} \omega & \mathbf{D}_t(i, j) > \tau_d \\ 0 & \text{otherwise} \end{cases}$ ▷ Instantaneous increment map
 - 4: $\mathbf{K}_t \leftarrow \min(\mathbf{K}_{t-1} + \mathbf{\Delta}_t, K_{\text{max}})$ ▷ Accumulate and cap
 - 5: $\mathbf{K}_t \leftarrow \max(\mathbf{K}_t - \delta, 0)$ ▷ Temporal decay
 - 6: $\mathbf{M}_t(i, j) \leftarrow \begin{cases} 255 & \mathbf{K}_t(i, j) \geq \tau_m \\ 0 & \text{otherwise} \end{cases}$ ▷ Threshold to binary mask
 - 7: **return** $\mathbf{M}_t, \mathbf{K}_t$
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Algorithm 3 Skip Module $\mathcal{S}$: Motion-Based Frame Gating

Require: Video stream $\{f_1, f_2, \dots\}$, motion detector $\mathcal{D}$, scale factor α , block size B , block active threshold τ

Ensure: Skip decisions $\{s_1, s_2, \dots\}$, where $s_t \in \{0, 1\}$

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1:  $f_{\text{prev}} \leftarrow \text{null}$  ▷ Initialize previous frame
2: for each incoming frame  $f_t$  do
    Step 1: Pre-process
3:   Downsample  $f_t$  by  $\alpha$  to obtain  $\tilde{f}_t$ 
4:   Let  $B_s \leftarrow B \cdot \alpha$  ▷ Effective block size on downsampled frame
5:   Pad  $\tilde{f}_t$  to be divisible by  $B_s$ 
    Step 2: Compute Foreground Mask
6:   if  $f_{\text{prev}} = \text{null}$  then ▷ No previous frame on first iteration
7:      $s_t \leftarrow 1$  ▷ Run inference on first frame
8:      $f_{\text{prev}} \leftarrow \tilde{f}_t$ 
9:   continue
10:  end if
11:   $\mathbf{M}_t \leftarrow \mathcal{D}(\tilde{f}_t, f_{\text{prev}})$  ▷  $\mathcal{D}$ : e.g., FrameDiff or AccMotionDet
12:  Partition  $\mathbf{M}_t$  into a grid of non-overlapping blocks of size  $B_s$ 
13:  Mark a block as active if its foreground pixel ratio  $> \tau$ 
    Step 3: Gating Decision
14:  if any active block exists then
15:     $s_t \leftarrow 1$  ▷ Motion detected, run DL inference
16:  else
17:     $s_t \leftarrow 0$  ▷ No motion, skip inference
18:  end if
19:   $f_{\text{prev}} \leftarrow \tilde{f}_t$  ▷ Update previous frame for next iteration
20:  yield  $s_t$ 
21: end for
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